

OSSP (Operating Systems And Systems Programming) - 25CS2104E

Practical 01

1A). Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that

1. Accept a Linux command as input.
2. Create a child process using fork).
3. Execute the command in the child process using an appropriate exec() system call.
4. Allow the parent process to wait for the child using wait ().
5. Display the Process ID (PID) of both parent and child processes.

1B). Using Linux terminal commands (uname, Iscpu, Isblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Part 1: C Program using fork(), exec(), and wait()

Step 1: Create a C file and write the code.

```
sahasra@Mac ~ % nano forkexec.c
```

```
UW PICO 5.09      File: forkexec.c      Modified
{
    printf("\nParent Process\n");
    printf("Parent PID : %d\n", getpid());

    wait(NULL);

    printf("Child process completed.\n");
}

return 0;
}
```

Step 2: Save the file

By pressing Ctrl+O —> Enter —> Ctrl+X

Step 3: Compile the program and Run the program

```
sahasra@Mac ~ % gcc forkexec.c -o forkexec
sahasra@Mac ~ % ./forkexec
Enter Linux command: 
```

```
Enter Linux command: ls

Parent Process
Parent PID : 3017

Child Process
Child PID : 3024
__pycache__                forkexec.c
<!DOCTYPE html>.html      from dataclasses import dataclass.py
app.py                     import streamlit as st.py
Applications               Internal.html
canteen-preorder          Java.java
CFAIPproject              Library
CFAIPProject.py           Movies
checkLeapYear(year) {.js  Music
const getGrade = student => {.js package-lock.json
default.profracw          Pictures
Desktop                   Project.html
DineTechJavaProject.java  ProjectSEM2.html
Documents                 Public
Downloads                 PyCharmMiscProject
fork_exec                 st.py
fork_exec.c               streamlit.py
forkexec
Child process completed.
```

Part 2: Operating System Commands

1. Display Operating System Information

```
sahasra@Mac ~ % uname -a
Darwin Mac25.5.0 Darwin Kernel Version 25.5.0: Tue Jun 9 22:26:22 PDT 2026; root:xnu-12377.121.10~1/RELEASE_ARM64_T8132 arm64
```

2. Display CPU Information

```
sahasra@Mac ~ % system_profiler SPHardwareDataType
2026-08-02 19:44:09.972 system_profiler[3044:487464] hw.cpubfamily: 0x6f5129ac
Hardware:

Hardware Overview:

Model Name: MacBook Air
Model Identifier: Mac16,12
Model Number: MC6T4HN/A
Chip: Apple M4
Total Number of Cores: 10 (4 Performance and 6 Efficiency)
Memory: 16 GB
System Firmware Version: 18000.121.3
OS Loader Version: 18000.121.3
Serial Number (system): LG9NJG793T
Hardware UUID: B2E2436B-6643-5C4C-ABCE-66448F86E00B
Provisioning UDID: 00008132-001928923A7A801C
Activation Lock Status: Disabled
```

3. Display Storage Information

```
[sahasra@Mac ~ % diskutil list
/dev/disk0 (internal, physical):
#:          TYPE NAME              SIZE      IDENTIFIER
0:          GUID_partition_scheme  *251.0 GB  disk0
1:          Apple_APFS_ISC Container disk1  524.3 MB  disk0s1
2:          Apple_APFS Container disk3  245.1 GB  disk0s2
3:          Apple_APFS_Recovery Container disk2  5.4 GB    disk0s3

/dev/disk3 (synthesized):
#:          TYPE NAME              SIZE      IDENTIFIER
0:          APFS Container Scheme -   +245.1 GB  disk3
                        Physical Store disk0s2
1:          APFS Volume Macintosh HD  12.6 GB    disk3s1
2:          APFS Snapshot com.apple.os.update-... 12.6 GB    disk3s1s1
3:          APFS Volume Preboot       9.0 GB     disk3s2
4:          APFS Volume Recovery      1.3 GB     disk3s3
5:          APFS Volume Data          108.1 GB   disk3s5
6:          APFS Volume VM            20.5 KB    disk3s6
```

4. Display Running Processes

```
sahasra@Mac ~ % ps
  PID TTY          TIME CMD
 3235 ttys001    0:00.02 -zsh
```

5. Display Live System Usage

```
sahasra@Mac ~ % top
```

```
Processes: 730 total, 3 running, 1 stuck, 726 sleeping, 3036 threads                                19:46:08
Load Avg: 3.44, 2.94, 2.79  CPU usage: 11.73% user, 6.43% sys, 81.83% idle
SharedLibs: 1135M resident, 217M data, 147M linkedit.
MemRegions: 0 total, 0B resident, 0B private, 2507M shared.
PhysMem: 15G used (3057M wired, 1055M compressor), 309M unused.
VM: 298T vsize, 6144M framework vsize, 0(0) swapins, 0(0) swapouts.
Networks: packets: 9505208/7909M in, 9250239/7724M out.
Disks: 1266987/23G read, 730945/11G written.

  PID   COMMAND      %CPU   TIME    #TH   #WQ   #PORT  MEM    PURG   CMPRS  PGRP  PPID  STATE
  798   avconference  48.1   12:14.53  48    8     1541+  336M   0B     1856K  798   1     stuck
  394   WindowServer  45.2   02:47:06  25/1   6     3262   418M   138M-  58M   394   1     running
  527   cameracaptur  24.5   02:50:51  18     4     349-   414M   0B     1184K  527   1     sleeping
   0    kernel_task   16.3   01:58:20  618/10 0      0      572M   0B     0B     0      0     running
 2971   VTEncoderXPC  11.2   02:11.03  5      2     119+   93M-   0B     912K   2971  1     sleeping
 3062   top           5.4    00:01.02  1/1    0      29     8384K+ 0B     0B     3062 2995  running
 3063   screencaptur  5.1    00:00.44  4      1     217    8976K  752K   0B     611   611   sleeping
 3000   com.apple.We  4.4    00:52.53  5      3     67     400M   0B     8080K  3000  1     sleeping
 2972   VTDecoderXPC  3.5    01:11.70  6      1     98     59M    0B     816K   2972  1     sleeping
 386   bluetoothd    2.8    00:54.53  12     3     430    11M    112K   2000K  386   1     sleeping
 412   coreaudiod    2.4    24:07.04  13     4     1269   27M    0B     7792K  412   1     sleeping
 530   com.apple.Dr  2.2    22:26.85  7      5     1347   30M    0B     3136K  530   1     sleeping
 1102   Safari        1.1    01:37.98  7      2     637    111M   133M   53M    1102  1     sleeping
 389   corebrightne  0.9    11:15.89  3      2     190    5200K  0B     1248K  389   1     sleeping
 804   AudioConvert  0.8    00:02.45  6      1     65     3968K  0B     608K   804   1     sleeping
 749   Spotify Help  0.7    12:28.99  18     2     216    120M   4768K  34M    734   734   sleeping
 369   locationd     0.6    00:37.89  6      3     303    10M    240K   2128K  369   1     sleeping
```